

ISA 444/5444: Business Forecasting

13: Baseline Forecasts with Exponential Smoothing Methods

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 Automated Scheduler for Office Hours

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Quick Refresher of Last Class

- ✓ Recognize the importance of baseline forecasts in time series forecasting
- ✓ Identify common baseline forecasts (Naive, Seasonal Naive, etc.)
- ✓ Implement [baseline models with StatsForecast](#)

Learning Objectives for Today's Class

- Understand the mathematical background for exponential smoothing methods, including:
 - Simple Exponential Smoothing (SES)
 - Double Exponential Smoothing (Holt)
 - Triple Exponential Smoothing (Holt-Winters)
- Implement these exponential smoothing methods in Python using StatsForecast

Exponential Smoothing Methods

Context: Our Previously Discussed Baseline Models

We have expressed our time series as: $y_1, y_2, \dots, y_T$.

Naive Forecast:

$$\hat{y}_{T+h|T} = y_T$$

Historical Average:

$$\hat{y}_{T+h|T} = \frac{1}{T} \sum_{t=1}^T y_t$$

Is there a more reasonable middle ground between these two extremes?

- Naive:
- Historical Average:
-

Simple Exponential Smoothing (SES)

Forecast equation:

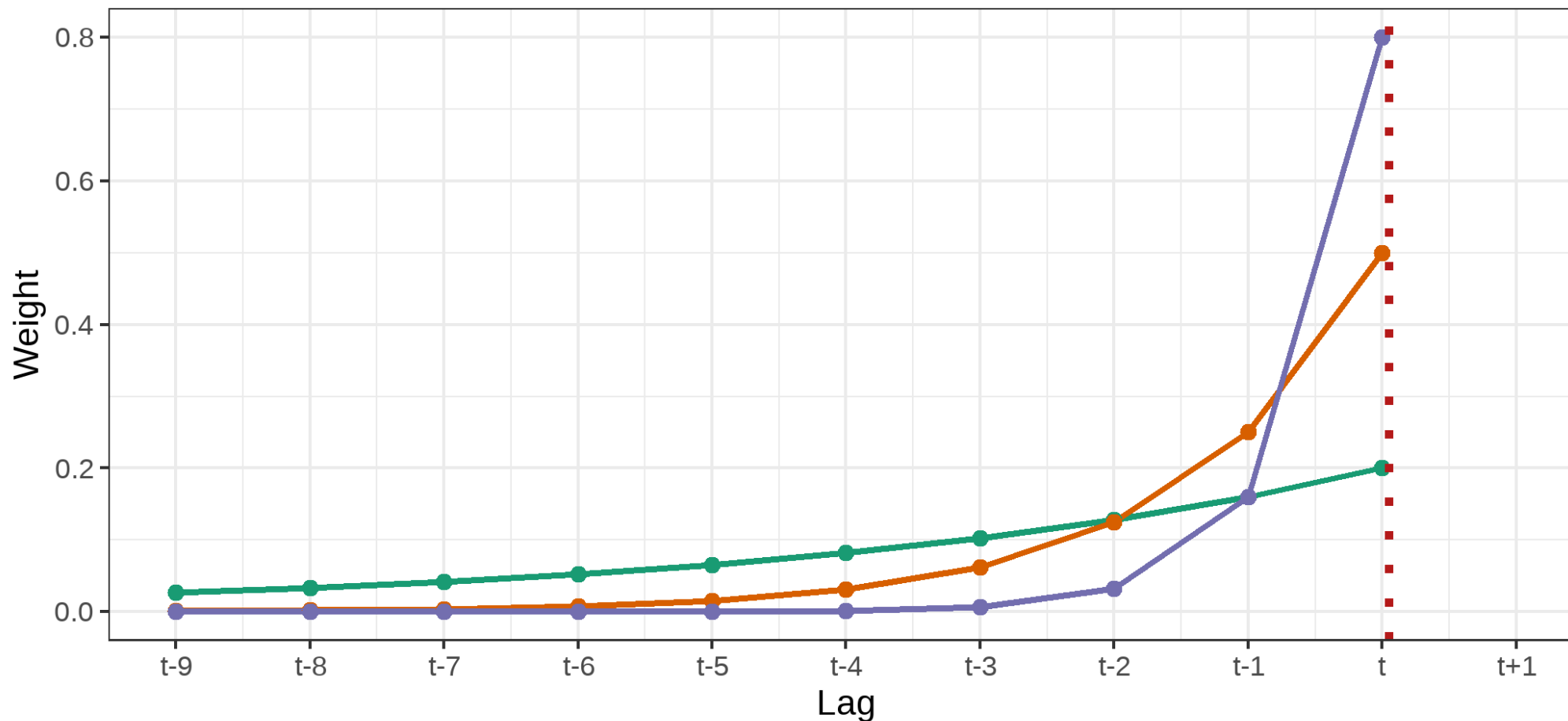
$$\hat{y}_{T+1|T} = \alpha y_T + \alpha(1 - \alpha)y_{T-1} + \alpha(1 - \alpha)^2 y_{T-2} + \alpha(1 - \alpha)^3 y_{T-3} + \cdots,$$

where $0 \leq \alpha \leq 1$ is the smoothing parameter.

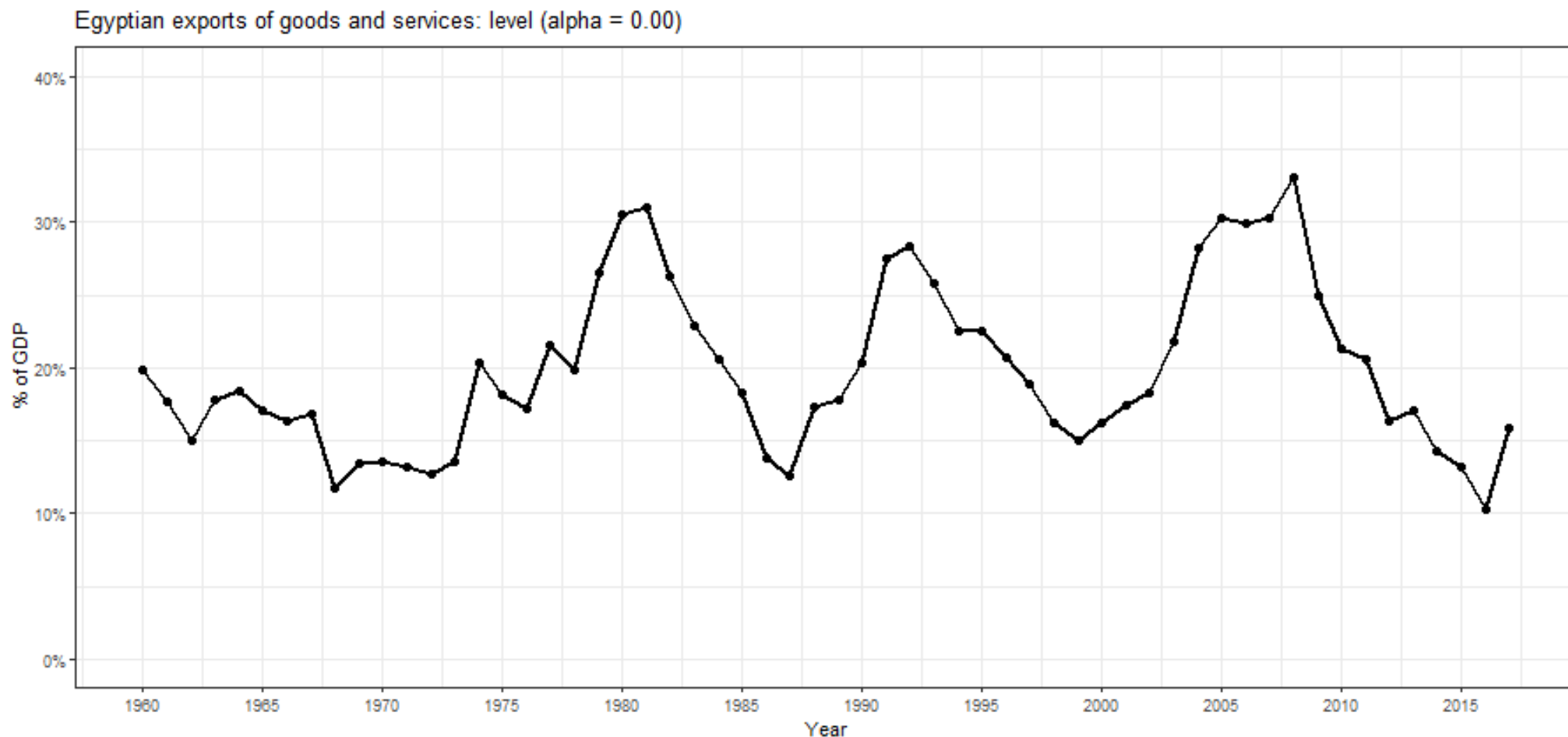
	Exponentially decaying weights assigned to observations for:			
Observation	$\alpha = 0.2$	$\alpha = 0.4$	$\alpha = 0.6$	$\alpha = 0.8$
y_T	0.2	0.4	0.6	0.8
y_{T-1}	0.16	0.24	0.24	0.16
y_{T-2}	0.128	0.144	0.096	0.032
y_{T-3}	0.1024	0.0864	0.0384	0.0064
y_{T-4}	$(0.2)(0.8)^4$	$(0.4)(0.6)^4$	$(0.6)(0.4)^4$	$(0.8)(0.2)^4$

Simple Exponential Smoothing (SES)

Weights as a Function of Time for the SES Model



Simple Exponential Smoothing (SES)



Simple Exponential Smoothing (SES)

Component Form:

Forecast equation	$\hat{y}_{t+h t} = \ell_t$	(mathematically equivalent to equation in Slide 6)
Smoothing equation	$\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$	

- ℓ_t is the level (or the smoothed value) of the series at time t
- SES is **computationally efficient**: $\hat{y}_{t+h|t} = \alpha y_t + (1 - \alpha)\hat{y}_{t|t-1}$
- The SES implementation assumes **no trend and no seasonality** since $\hat{y}_{T+h|T} = \ell_T$, for $h \geq 1$, and ℓ_T is a constant (last smoothed value).
- The SES model is equivalent to the ETS(A,N,N) model in the ETS framework (additive error, no trend, no seasonality).

Simple Exponential Smoothing (SES)

Weighted Average Form:

$$\hat{y}_{T+1|T} = \sum_{j=0}^{T-1} \alpha(1 - \alpha)^j y_{T-j} + (1 - \alpha)^T \ell_0$$

Smoothing Parameters in SES

- There are two parameters for the SES model: α , and ℓ_0 .
- These parameters are either **estimated** or **set** by the user.
- The [StatsForecast](#) library provides two implementation of SES:
 - `SimpleExponentialSmoothing`, where you can set the **alpha** parameter. It uses the first value of the time series as the initial level (ℓ_0); or
 - `SimpleExponentialSmoothingOptimized`, which finds the optimal α parameter by minimizing the sum of squared errors.
- Please refer to the [SimpleExponentialSmoothing Tutorial](#) and the [SimpleExponentialSmoothingOptimized Tutorial](#) for more details on the Nixtla Implementation.

Class Activity

Description	Relevant Info	Your Solution	Validate [Out of Class]
• In the beginning of the semester, I noted that the <i>StatsForecast</i> Python implementation focuses on forecasting at scale, while traditional R implementations focus on model explanation and interpretation. • The SimpleExponentialSmoothingOptimized model in StatsForecast is a good example of this, where the model is optimized to find the best α parameter but it does not directly provide the optimized α value. • Your task is to find the optimized α value from the reported fitted values from their example. • Note that the purpose of this activity is not the mathematical exercise, but to understand the model's output and to be able to explain it to your future colleagues.			

Interesting Interpretations of the SES Function

By rearranging the forecast equation, we can express it as:

$$\begin{aligned} \underbrace{\hat{y}_{t+1|t}}_{\text{Forecast for next time period}} &= \alpha y_t + (1 - \alpha) \hat{y}_{t|t-1} \\ &= \underbrace{\hat{y}_{t|t-1}}_{\text{Previous estimate}} + \underbrace{\alpha}_{\text{Smoothing parameter}} \underbrace{(y_t - \hat{y}_{t|t-1})}_{\text{Forecast error}} \\ &= \underbrace{\hat{y}_{t|t-1}}_{\text{Lag of level}} + \underbrace{\alpha}_{\text{Smoothing parameter}} \underbrace{(y_t - \hat{y}_{t|t-1})}_{\text{Remainder}} \end{aligned}$$

- The **forecast for next time period** is a weighted sum of the **actual observation** and **previous forecast**.
- The **smoothing parameter** α controls how much weight is given to the new observation.
- The **forecast error** $(y_t - \hat{y}_{t|t-1})$ captures how much the previous estimate deviates from the actual.
- The **previous estimate** $\hat{y}_{t|t-1}$ represents the forecast from the prior time step.

Double Exponential Smoothing (Holt)

- The SES model is suitable for time series with no trend or seasonality.
- The Holt model extends SES to time series with a **local (not constant) linear trend**, but no seasonality.
- The Holt model uses two smoothing parameters: α for the level and β^* for the trend.
- The Holt model is equivalent to the ETS(A,A,N) model in the ETS framework (additive error, additive trend, no seasonality).
- The Holt model is implemented in the [StatsForecast](#) library as `Holt`.

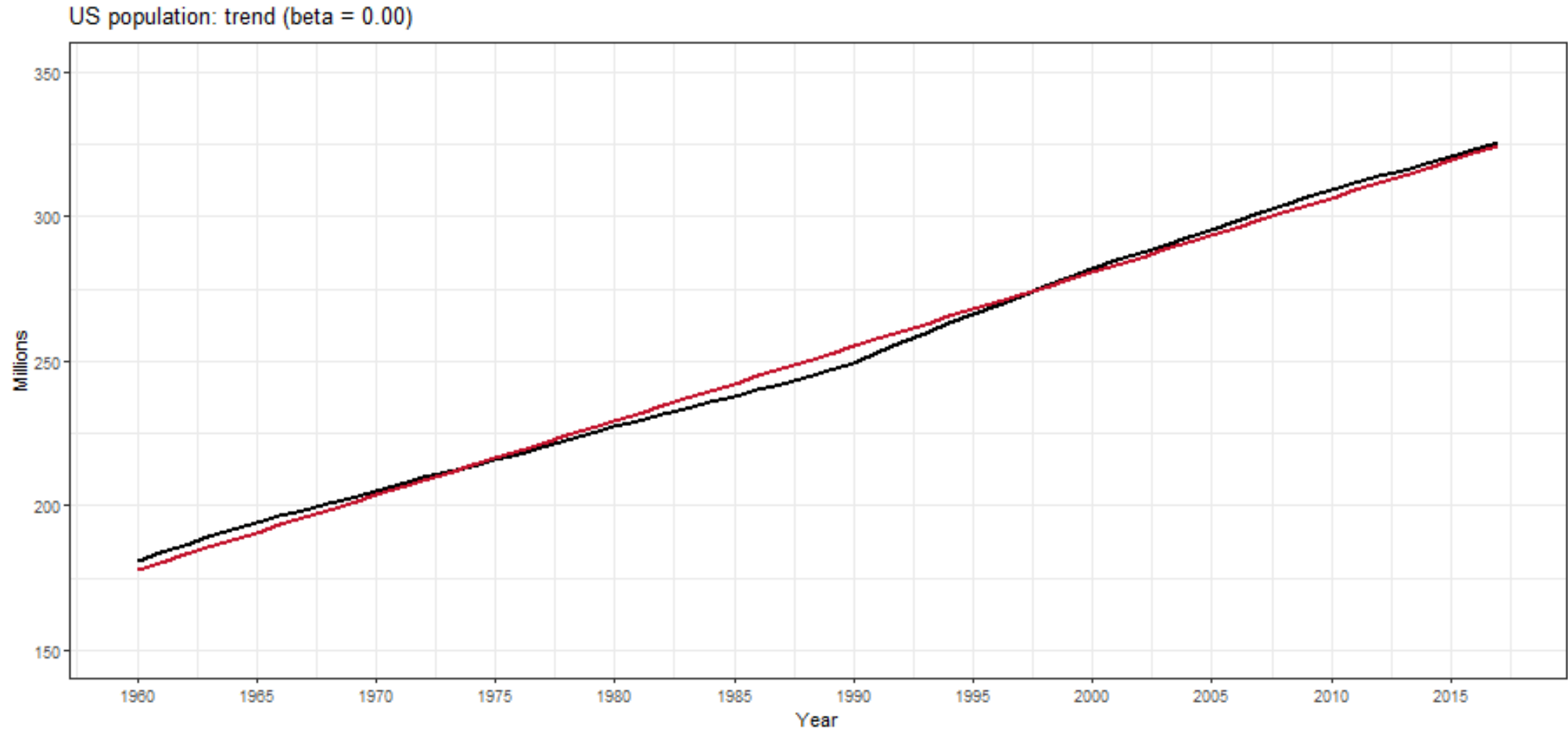
Double Exponential Smoothing (Holt)

$$\underbrace{\text{Forecast}}_{\text{Predict future value}} \quad \underbrace{\hat{y}_{t+h} | t}_{\text{Forecast for } t+h} = \underbrace{\ell_t}_{\text{Level (Base Value)}} + h \underbrace{b_t}_{\text{Trend (Growth Rate)}}$$

$$\underbrace{\text{Level}}_{\text{Smoothed estimate of series}} \quad \underbrace{\ell_t}_{\text{New level estimate}} = \underbrace{\alpha y_t}_{\text{Weight on new observation}} + \underbrace{(1 - \alpha)(\ell_{t-1} + b_{t-1})}_{\text{Weight on past level + trend}}$$

$$\underbrace{\text{Trend}}_{\text{Smoothed trend estimate}} \quad \underbrace{b_t}_{\text{New trend estimate}} = \underbrace{\beta^*(\ell_t - \ell_{t-1})}_{\text{New level change weighted by } \beta^*} + \underbrace{(1 - \beta^*)b_{t-1}}_{\text{Past trend estimate weighted by } (1 - \beta^*)}$$

Double Exponential Smoothing (Holt)



Triple Exponential Smoothing (Holt-Winters)

- The Holt-Winters model extends Holt to time series with **seasonality**.
- The Holt-Winters model uses three smoothing parameters: α for the level, β^* for the trend, and γ for the seasonality.
- There are two versions of the Holt-Winters model:
 - **Additive Seasonality**: The seasonal component is added to the level and trend.
 - **Multiplicative Seasonality**: The seasonal component is multiplied by the level and trend.
- The Holt-Winters model is implemented in the [StatsForecast](#) library as **HoltWinters**.

Triple Exponential Smoothing (Holt-Winters [Add])

$$\underbrace{\hat{y}_{t+h} | t}_{\text{Forecast for } t+h} = \underbrace{\ell_t}_{\text{Level (Base Value)}} + h \underbrace{b_t}_{\text{Trend (Growth Rate)}} + \underbrace{s_{t+h-m(k+1)}}_{\text{Seasonal Effect}}$$

$$\underbrace{\ell_t}_{\text{Smoothed estimate of level}} = \underbrace{\alpha(y_t - s_{t-m})}_{\text{Weight on seasonally adjusted observation}} + \underbrace{(1 - \alpha)(\ell_{t-1} + b_{t-1})}_{\text{Weight on past level + trend}}$$

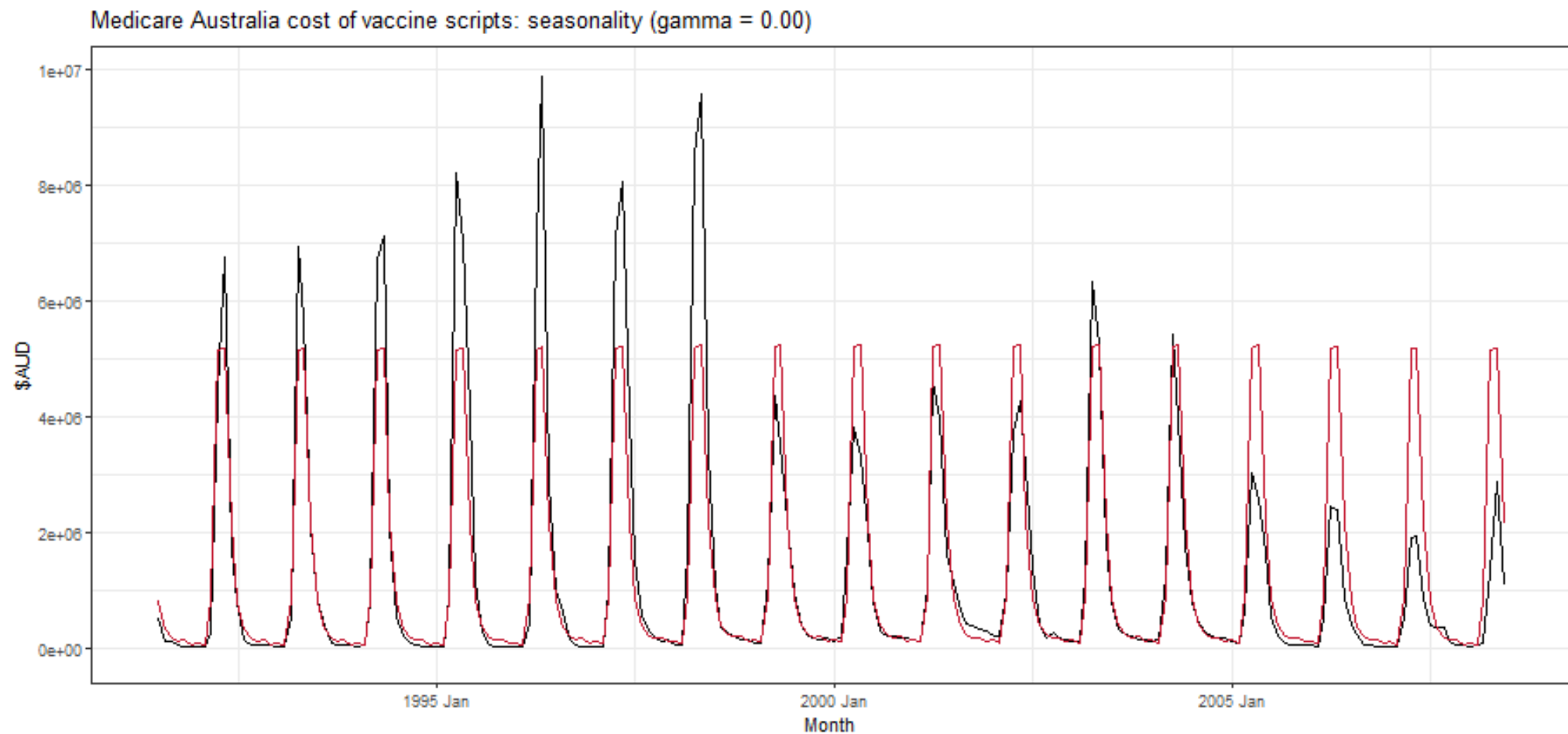
$$\underbrace{b_t}_{\text{Smoothed estimate of trend}} = \underbrace{\beta^*(\ell_t - \ell_{t-1})}_{\text{New level change weighted by } \beta^*} + \underbrace{(1 - \beta^*)b_{t-1}}_{\text{Past trend estimate weighted by } (1 - \beta^*)}$$

$$\underbrace{s_t}_{\text{Smoothed estimate of seasonality}} = \underbrace{\gamma(y_t - \ell_{t-1} - b_{t-1})}_{\text{Weight on new seasonal effect}} + \underbrace{(1 - \gamma)s_{t-m}}_{\text{Weight on past seasonal component}}$$

Triple Exponential Smoothing (Holt-Winters [Add])

- Seasonal component is usually expressed as $s_t = \gamma^*(y_t - \ell_t) + (1 - \gamma^*)s_{t-m}$.
- Substitute in for ℓ_t : $s_t = \gamma^*(1 - \alpha)(y_t - \ell_{t-1} - b_{t-1}) + [1 - \gamma^*(1 - \alpha)]s_{t-m}$
- We set $\gamma = \gamma^*(1 - \alpha)$.
- The usual parameter restriction is $0 \leq \gamma^* \leq 1$, which translates to $0 \leq \gamma \leq (1 - \alpha)$.

Triple Exponential Smoothing (Holt-Winters [Add])



Triple Exponential Smoothing (Holt-Winters [Mult])

Seasonal variations change in proportion to the level of the series.

$$\underbrace{\hat{y}_{t+h|t}}_{\text{Forecast at time } t+h} = \underbrace{(\ell_t + hb_t)}_{\substack{\text{Level + Trend} \\ \text{(Growth Adjusted)}}} \times \underbrace{s_{t+h-m(k+1)}}_{\substack{\text{Seasonal} \\ \text{Effect}}}$$

$$\underbrace{\ell_t}_{\substack{\text{Smoothed estimate} \\ \text{of level}}} = \underbrace{\alpha \frac{y_t}{s_{t-m}}}_{\substack{\text{Seasonally adjusted} \\ \text{new observation}}} + \underbrace{(1 - \alpha)(\ell_{t-1} + b_{t-1})}_{\substack{\text{Weight on past} \\ \text{level + trend}}}$$

$$\underbrace{b_t}_{\substack{\text{Smoothed estimate} \\ \text{of trend}}} = \underbrace{\beta^*(\ell_t - \ell_{t-1})}_{\substack{\text{New level change} \\ \text{weighted by } \beta^*}} + \underbrace{(1 - \beta^*)b_{t-1}}_{\substack{\text{Past trend estimate} \\ \text{weighted by } (1-\beta^*)}}$$

$$\underbrace{s_t}_{\substack{\text{Smoothed estimate} \\ \text{of seasonality}}} = \underbrace{\gamma \frac{y_t}{(\ell_{t-1} + b_{t-1})}}_{\substack{\text{New seasonal adjustment} \\ \text{weighted by } \gamma}} + \underbrace{(1 - \gamma)s_{t-m}}_{\substack{\text{Past seasonal component} \\ \text{weighted by } (1-\gamma)}}$$

Recap

Summary of Main Points

By now, you should be able to do the following:

- Understand the mathematical background for exponential smoothing methods, including:
 - Simple Exponential Smoothing (SES)
 - Double Exponential Smoothing (Holt)
 - Triple Exponential Smoothing (Holt-Winters)
- Implement these exponential smoothing methods in Python using StatsForecast



Review and Clarification



- **Class Notes:** Take some time to revisit your class notes for key insights and concepts.
- **Zoom Recording:** The recording of today's class will be made available on Canvas approximately 3-4 hours after the session ends.
- **Questions:** Please don't hesitate to ask for clarification on any topics discussed in class. It's crucial not to let questions accumulate.